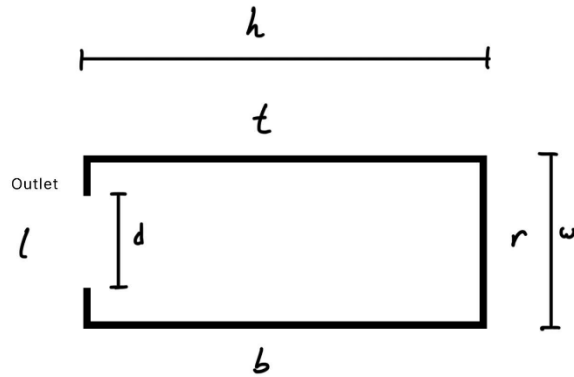


Problem definition

Short description of the flow problem your group wants to work on.

We want to model the distribution of heat and velocity of escaping gases from a smoke grenade. Our goal is to experiment with the stoichiometric ratios of potassium nitrate and sugar and how they affect the temperature of escaping gas during combustion. We will optimize for the most amount of potassium carbonate left after the reaction, which is the main solid particulate that creates smoke, while keeping the temperature of escaping gases below 891 degrees Celsius to avoid vaporizing the smoke particles.



Sketch

A simple sketch of your working domain (dimensions, inlet, outlet, walls, baffles, ...)

Smoke Grenade opening – Outlet (d): Variable velocity dependent on generated pressure from reaction

Smoke Grenade walls (l, t, r, b) – wall: no-slip BC

Right side wall – Neumann: no heat transfer across boundary

Other walls – Neumann: Prescribed heat flux based on convection estimates

Transport equations

What transport equations do you need to solve? What variables?

Equations:

continuity, momentum (Navier-Stokes), species transport, reaction equations

Variables:

Species mass and mole fractions, pressure, velocity, temperature, density, volume

Research questions

What are the research questions you'd like to answer in this assignment?

- 1. What is the effect of the stoichiometric composition on resulting amount of potassium carbonate?*
- 2. What is the effect of the stoichiometric composition on temperature distribution?*